Infosys System Engineer: Mock Test — Paper 3

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: All P are Q. All Q are R. Some R are S. No S is T.
Conclusions: I. All P are R. II. Some Q are S. III. No P is T. IV. Some R are not T.
(a) Only I follows (b) I and IV follow (c) I, II and IV follow (d) All follow

Q2. Statements: All coins are metals. No metal is wood. Some woods are furniture.
Conclusions: I. No coin is wood. II. Some furniture is not metal.
(a) Only I follows (b) Only II follows (c) Both I and II follow (d) Neither follows

Q3. *Data Sufficiency.* What is the two-digit number N?
I. The sum of its digits is 9.
II. The number obtained by reversing its digits is 27 less than N.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Each statement alone is sufficient
(e) Even both statements together are not sufficient

Q4. *Data Sufficiency.* Is the integer x even?
I. x/2 is an integer.
II. x^2 is even.
(a) Statement I alone is sufficient, but II alone is not
(b) Statement II alone is sufficient, but I alone is not
(c) Both statements together are sufficient, but neither alone is
(d) Each statement alone is sufficient
(e) Even both statements together are not sufficient

Q5. In a certain code, TRAIN is written as 23145, GAME is written as 6178, and MEAT is written as 7812. What is the code for GATE?
(a) 6128 (b) 6218 (c) 6182 (d) 8126

Q6. In a code: "bright light shines" = "mo ta pu", "shines through night" = "pu ri ke", "night bright stars" = "ke mo lo", "stars through galaxy" = "lo ri zu". What is the code for "night light galaxy"?

(a) ke ta zu (b) ke ta ri (c) mo ta zu (d) ke lo zu

Q7. A cuboid measuring 5 × 4 × 3 units is painted on all its outer surfaces and then cut into unit cubes. How many unit cubes have exactly 2 painted faces?

(a) 24 (b) 22 (c) 20 (d) 26

Q8. How many triangles are there in total in a complete five-pointed star (pentagram) figure?

(a) 5 (b) 10 (c) 35 (d) 15

Q9. Introducing a man, a woman says, "His wife is the only daughter-in-law of my father." How is the woman related to the man's wife?

(a) Sister-in-law (b) Sister (c) Mother (d) Aunt

Q10. If A % B means A is the mother of B, and A * B means A is the father of B, which expression shows that C is the paternal grandmother of D?

(a) C % E * D (b) C * E % D (c) E % C * D (d) D % E * C

Q11. A man walks 12 m North, then 9 m East, then 8 m South, then 6 m West. How far is he from the starting point, and in which general direction?

(a) 5 m, North-East (b) 5 m, North-West (c) 7 m, North-East (d) 5 m, South-East

Q12. A cyclist starts facing East. He turns 135° clockwise, then 90° anticlockwise, then 180° clockwise. Which direction is he facing now?

(a) North-West (b) South-East (c) South-West (d) North-East

Q13. Find the next term: 7, 26, 63, 124, 215, ?

(a) 342 (b) 344 (c) 341 (d) 343

Q14. Find the next term: 4, 5, 9, 18, 34, ?

(a) 55 (b) 59 (c) 63 (d) 57

Q15. Find the next term: B, D, G, K, P, ?

(a) V (b) U (c) T (d) W

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?

(a) 1260 (b) 2520 (c) 630 (d) 420

Q17. A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?

(a) 658 (b) 686 (c) 700 (d) 623

Q18. Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?

(a) 4/11 (b) 7/11 (c) 2/5 (d) 1/2

Q19. A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?

(a) 4/33 (b) 3/33 (c) 5/33 (d) 4/29

Q20. A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ₹40/kg and ₹60/kg) works out to ₹48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.

(a) 38.9% (b) 35% (c) 41.7% (d) 33.3%

Q21. Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?

(a) 4 hours (b) 3 hours (c) 5 hours (d) 6 hours

Q22. In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?

(a) 190 m (b) 200 m (c) 180 m (d) 210 m

Q23. An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.

(a) 12 kg (b) 10 kg (c) 15 kg (d) 8 kg

Q24. ₹8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.

(a) ₹20 (b) ₹40 (c) ₹80 (d) ₹100

Q25. In what time will ₹8,000 amount to ₹9,261 at 5% per annum, compounded annually?

(a) 2 years (b) 3 years (c) 4 years (d) 3.5 years

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *Sokware effort estimation has resisted four decades of methodological refinement, and the reason is not, as oken supposed, a lack of adequate techniques. Function point analysis, story point velocity tracking and Monte Carlo simulation are all, in controlled retrospective studies, reasonably well calibrated. The persistent gap between estimated and actual delivery dates arises instead from a structural asymmetry: the costs of underestimating are diffuse and delayed, distributed across a project's eventual overrun, whereas the costs of overestimating are immediate and attributable, since a padded estimate can lose a bid or provoke management scrutiny before the work even begins. Estimators therefore face an incentive gradient that has nothing to do with statistical accuracy and everything to do with organisational consequence, and this gradient reliably pushes estimates toward optimism regardless of how sophisticated the underlying model is. Attempts to correct this by adding explicit contingency buffers frequently fail for a related reason: once a buffer becomes visible in a schedule, it is treated by stakeholders as slack to be reclaimed rather than risk to be reserved, a phenomenon sometimes called the student syndrome, in reference to the tendency to begin a task only as its deadline approaches.*

Q26. The passage's central claim is that inaccurate software estimates stem primarily from:

(a) Inadequate estimation techniques that fail to model complexity

(b) An organisational incentive structure that rewards optimistic estimates over accurate ones

(c) The inherent unpredictability of software requirements

(d) A lack of contingency buffers in project schedules

Q27. According to the passage, the "student syndrome" refers to:
(a) Estimators lacking sufficient training in statistical methods
(b) Visible schedule buffers being treated as reclaimable slack rather than reserved risk
(c) Junior developers underestimating their own task durations
(d) Students generally being poor at time management

Q28. Which of the following best captures the "structural asymmetry" the author describes?
(a) Overestimation carries immediate, visible costs, while underestimation's costs are delayed and diffuse
(b) Function point analysis is less accurate than story point velocity
(c) Management scrutinises all estimates equally regardless of direction
(d) Contingency buffers eliminate the risk of underestimation

Q29. The author implies that adding explicit contingency buffers to schedules:
(a) Always improves estimation accuracy
(b) Can be counterproductive once stakeholders perceive the buffer as available slack
(c) Is the only reliable solution to the estimation problem
(d) Has no effect on stakeholder behaviour

Q30. Which statement would the author most likely dispute?
(a) Estimation techniques themselves are the primary obstacle to accurate forecasting
(b) Incentives shape estimation behaviour independent of technique sophistication
(c) A padded estimate can cost a bid before work begins
(d) Underestimation costs tend to surface only over the course of a project

Sentence Correction (Q31–35).

Q31. Not only was the deployment delayed by unforeseen dependencies, **it also required** a complete rollback of the previous release.
(a) No improvement (b) but it also required (c) also it required (d) but also it required

Q32. The findings, **which the researchers were confident in it**, contradicted the earlier hypothesis.
(a) No improvement (b) which the researchers were confident about it (c) in which the researchers were confident (d) that the researchers were confident of it

Q33. Having reviewed the architecture for three days, **the decision was made** to migrate to microservices.
(a) No improvement (b) the team decided (c) it was decided by the team (d) the migration decision was made

Q34. The new algorithm is **more efficient as compared to** the previous version.
(a) No improvement (b) more efficient than (c) more efficient in comparison of (d) as more efficient than

Q35. Having been informed of the outage, **immediate action was taken by the on-call engineer**.
(a) No improvement (b) the on-call engineer took immediate action (c) it was immediately actioned by the on-call engineer (d) immediate action, taken by the on-call engineer

Para-jumbles (Q36–38).

Q36.
P: Yet the very flexibility that made the format popular also made it difficult to validate programmatically.
Q: JSON emerged as a lightweight alternative to XML for data interchange between web services.
R: Schema languages such as JSON Schema were developed specifically to address this gap.
S: Unlike XML, it dispensed with the rigid schema definitions many developers found cumbersome.
(a) QSPR (b) QPSR (c) SQPR (d) QSRP

Q37.

P: This is because correlation alone cannot distinguish a cause from a shared underlying factor.

Q: Two variables can move together closely across an entire dataset without either one causing the other.

R: Researchers therefore rely on controlled experiments to isolate genuine causal effects.

S: Ice cream sales and drowning incidents, for instance, both rise in summer, driven by heat rather than by each other.

(a) QSPR (b) QPSR (c) SQPR (d) QPRS

Q38.

P: The resulting backlog of unresolved tickets grew faster than the team could triage it.

Q: The support team initially handled every incoming ticket in the order it arrived.

R: A priority-based queue was introduced only after complaints about critical issues being delayed behind trivial ones.

S: This approach ignored the fact that some issues were far more urgent than others.

(a) QSPR (b) QPSR (c) SQPR (d) PQSR

Vocabulary, Synonyms and Antonyms (Q39–45).

Q39. The board's decision to proceed despite the warnings was widely seen as **obstinate** rather than courageous. "Obstinate" means:

(a) flexible (b) stubbornly unwilling to change (c) courageous (d) uncertain

Q40. His **perfunctory** apology did little to repair the damaged trust. "Perfunctory" means:

(a) heartfelt (b) done as a mere formality, without real effort (c) delayed (d) elaborate

Q41. Fill in the blank: The consultant's report was criticised for being _____, offering broad generalisations instead of concrete recommendations.

(a) meticulous (b) vague (c) precise (d) exhaustive

Q42. Synonym of **INSCRUTABLE**:

(a) transparent (b) unfathomable (c) predictable (d) simple

Q43. Synonym of **PROFLIGATE**:

(a) frugal (b) wasteful (c) careful (d) modest

Q44. Antonym of **ASSIDUOUS**:

(a) diligent (b) careful (c) negligent (d) thorough

Q45. Antonym of **EPHEMERAL**:

(a) fleeting (b) permanent (c) brief (d) transient

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed?

```
int f(int n) {
    if (n <= 0) return 0;
    if (n == 1) return 1;
    return f(n-1) + f(n-2) + n;
}
print(f(6))
```

(a) 40 (b) 46 (c) 52 (d) 44

Q47. What is printed?

```
x = 37; mask = 0;
for i = 0 to 5
    if (((x >> i) & 1) == 1)
        mask = mask | (1 << (5 - i))
    end if
end for
print(mask)
```

(a) 37 (b) 41 (c) 45 (d) 39

Q48. What is printed?

```
s = 0; i = 1
while (i <= 6)
    j = 1
    while (j <= i)
        if (j % 2 == 0)
            j = j + 1
            continue
        end if
        s = s + (i * j)
        j = j + 1
    end while
    i = i + 1
end while
print(s)
```

(a) 130 (b) 124 (c) 136 (d) 118

Q49. A stack is used (LIFO). What is printed?

```
push(4); push(7);
a = pop(); b = pop();
push(a - b);
push(10);
c = pop(); d = pop();
push(c + d);
push(3);
e = pop(); f = pop();
print(e * f)
```

(a) 39 (b) 33 (c) 42 (d) 30

Q50. What is the final content of `arr`? (0-indexed; each statement uses the array's state at that point)

```
arr = [10, 20, 30, 40, 50, 60]
i = 0; j = 5
while (i < j)
    temp = arr[i]
    arr[i] = arr[j]
    arr[j] = temp
    i = i + 1
    j = j - 2
end while
print(arr)
```

(a) [60, 40, 30, 20, 50, 10] (b) [60, 20, 30, 40, 50, 10] (c) [10, 40, 30, 20, 50, 60] (d) [60, 40, 50, 20, 30, 10]

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* A figure is made of 3 unit squares arranged in an L-shape (one square with a second attached to its right, and a third attached below the first), using 10 matchsticks in total. What is the minimum number of sticks to remove so that no complete square remains?

(a) 1 (b) 2 (c) 3 (d) 4

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 5, R = 3, B = 7, L = 2, Centre = 19

Wheel 2: T = 4, R = 6, B = 9, L = 3, Centre = 63

Wheel 3: T = 8, R = 5, B = 6, L = 4, Centre = ?

(a) 54 (b) 58 (c) 62 (d) 50

Q53. *Matrix logic.* Find the missing number.

4	9	25
9	25	49
25	49	?

(a) 100 (b) 121 (c) 144 (d) 81

Q54. *Pattern.* Find the next term: 6, 24, 60, 120, 210, ?

(a) 320 (b) 330 (c) 336 (d) 342

SECTION 6: English Grammar (5 Q, 10 min)

Q55. A number of employees _____absent, but the majority of the workforce _____present today.

(a) is, is (b) are, is (c) is, are (d) are, are

Q56. By the time she retires next year, she _____for this company for over two decades.

(a) will work (b) will have worked (c) will have been working (d) has worked

Q57. The proposal was drafted _____close consultation _____the stakeholders and was later submitted _____review.

(a) in, with, for (b) with, to, in (c) in, to, for (d) by, with, to

Q58. *Error spoṅng.* "The committee not only approved the budget (A) / but also the timeline for implementation (B) / were revised (C) / to reflect the new priorities. (D)"

(a) A (b) B (c) C (d) D

Q59. _____honesty is _____best policy, though _____history is full of examples where _____ dishonest have prospered.

(a) An, the, -, the (b) The, the, -, the (c) -, the, -, the (d) An, a, the, an

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write a short technical essay/paragraph of 150–180 words.*

Explain, in your own words, the trade-offs between vertical scaling (upgrading a single server's capacity) and horizontal scaling (adding more servers) of a server infrastructure. Recommend which approach a mid-sized e-commerce company experiencing rapid, unpredictable traffic growth should adopt, and justify your recommendation.

Evaluation rubric (10 marks): Technical accuracy of the comparison (3) | Clarity of the recommendation and justification (3) | Structure and coherence (2) | Grammar and vocabulary (1) | Word limit adherence (1)

ANSWER KEY WITH EXPLANATIONS

Section 1: Reasoning

1. **(b)** Since $P \subseteq Q \subseteq R$, "All P are R" (I) follows directly. Some R are S, and no S is T, so those particular R-elements are not T — "Some R are not T" (IV) follows. Neither II nor III can be derived, since we don't know how P or Q specifically relates to the R-elements that happen to be S.

2. **(c)** Coins$\subseteq$metals, and metals$\cap$wood=$\varnothing$, so coins$\cap$wood=$\varnothing$ — I follows. Some woods are furniture, and since no metal is wood, those wood-furniture items are not metal — II follows.

3. **(c)** II alone: N − reversed = 9(tens digit − units digit) = 27, so tens − units = 3 — multiple numbers fit (41, 52, 63, 74, 85, 96), insufficient alone. I alone: digit sum = 9 also fits multiple numbers, insufficient alone. Together: tens − units = 3 and tens + units = 9 gives tens = 6, units = 3, so N = 63 — unique.

4. **(d)** I alone: x/2 being an integer directly means x is even — sufficient. II alone: x^2 even is only possible if x itself is even (an odd number squared is always odd) — sufficient. Each statement alone is sufficient.

5. **(a)** From TRAIN=23145: T=2,R=3,A=1,I=4,N=5. From GAME=6178: G=6,A=1,M=7,E=8. From MEAT=7812: confirms M=7,E=8,A=1,T=2. So GATE = G(6)A(1)T(2)E(8) = **6128**.

6. **(a)** Comparing statement 1 & 2, the common word "shines" gets the common code "pu". Comparing 2 & 3, "night" gets the common code "ke". Comparing 1 & 3, "bright" gets the common code "mo". This leaves light=ta, stars=lo, through=ri, galaxy=zu (verified consistent with statement 4). So "night light galaxy" = **ke ta zu**.

7. **(a)** For a cuboid a×b×c, the number of unit cubes with exactly 2 painted faces = 4[(a−2)+(b−2)+(c−2)]. With a=5,b=4,c=3: 4[3+2+1] = 4×6 = **24**.

8. **(c)** A complete five-pointed star (pentagram) figure contains **35** triangles in total (5 small tip triangles, 5 medium, 5 larger, and combinations up to the whole star) — a standard, well-documented result for this figure.

9. **(a)** The man's wife is the only daughter-in-law of the woman's father, meaning the woman's father has only one son — the man himself. So the man is the woman's brother, making the woman the **sister-in-law** of the man's wife.

10. **(a)** C % E: C is mother of E. E * D: E is father of D. So C is the mother of D's father, i.e., C is D's **paternal grandmother**.

11. **(a)** Net North-South displacement = 12−8 = 4 m North; net East-West = 9−6 = 3 m East. Distance = $\sqrt{(3^2+4^2)} = \sqrt{25}$ = **5 m**, in the general **North-East** direction.

12. **(a)** Using bearings (clockwise from North, East=90°): 90°+135°=225°(SW); 225°−90°=135°(SE); 135°+180°=315°, which is **North-West**.

13. **(a)** Each term is n^3-1 for n=2,3,4,5,6: 7,26,63,124,215; next (n=7): 343−1 = **342**.

14. **(b)** Differences are 1,4,9,16 (perfect squares); next difference is 25: 34+25 = **59**.

15. **(a)** Differences between consecutive letters (by position) increase by 1 each time: +2,+3,+4,+5, next +6: P(16)+6 = 22 = **V**.

Section 2: Quant

16. **(a)** Choose which of the 3 students gets 4 books: 3 ways. Choose that student's 4 books: C(8,4)=70. Split the remaining 4 books between the other two specific students, 2 each: C(4,2)=6. Total = 3×70×6 = **1,260**.

17. **(a)** Valid (men,women) splits with men>women, total=6: (6,0): C(7,6)×C(6,0)=7. (5,1): C(7,5)×C(6,1)=21×6=126. (4,2): C(7,4)×C(6,2)=35×15=525. Total = 7+126+525 = **658**. (Cross-check: adding the equal split (3,3)=700 and women>men splits=358 gives 658+700+358=1,716=C(13,6), confirming the count.)

18. **(a)** P(Red|A)=4/10, P(Red|B)=7/10, each bag chosen with probability 1/2. P(Red) = ½×0.4 + ½×0.7 = 0.55. By Bayes' theorem, P(A|Red) = (½×0.4)/0.55 = 0.2/0.55 = **4/11**.

19. **(a)** Favourable = C(5,2)×C(4,2) = 10×6 = 60. Total ways = C(12,4) = 495. Probability = 60/495 = **4/33**.

20. **(a)** Marked-up SP for a claimed 1 kg = 48×1.25 = ₹60. True cost of the 900 g actually given = 0.9×48 = ₹43.2. Profit = 60−43.2 = ₹16.8 on a true cost of ₹43.2, giving 16.8/43.2×100 ≈ **38.9%**.

21. **(a)** Combined rate (A+B−leak) = 1/12+1/15−1/20 = 5/60+4/60−3/60 = 1/10 per hour. In 4 hours, 4/10 = 2/5 of the tank fills, leaving 3/5. A+B's combined rate (no leak) = 1/12+1/15 = 3/20 per hour. Time for remaining 3/5 = (3/5)÷(3/20) = **4 hours**.

22. **(a)** Speed ratio A:B = 1000:900 = 10:9 (A beats B by 100 m in 1000 m). Similarly B:C = 10:9. So A:C = (10/9)×(10/9) = 100:81. When A covers 1000 m, C covers (81/100)×1000 = 810 m. A beats C by 1000−810 = **190 m**.

23. **(a)** Copper in original alloy = 0.6×20 = 12 kg; zinc = 8 kg (unchanged after adding pure copper). For 75% copper (25% zinc): 8 = 0.25×(20+x), so 20+x = 32, x = **12 kg**.

24. **(a)** Half-yearly compounding: 2 periods at 5% each. Amount = 8000×1.05×1.05 = 8,820, so CI = ₹820. SI = 8000×10%×1 = ₹800. Difference = 820−800 = **₹20**.

25. **(b)** 9,261/8,000 = 1.157625 = 1.05^3 (since 1.05^2=1.1025, ×1.05=1.157625). So the time is **3 years**.

Section 3: Verbal

26. **(b)** 27. **(b)** 28. **(a)** 29. **(b)** 30. **(a)**

31. **(b)** "Not only...but" needs "but" to complete the correlative pair; "but it also required" is the standard, grammatical form.

32. **(c)** The redundant pronoun "it" must be dropped, and the preposition placed before the relative pronoun: "in which the researchers were confident".

33. **(b)** "Having reviewed..." is a dangling modifier — it must describe the people who did the reviewing, so the sentence needs an explicit human subject: "the team decided".

34. **(b)** The correct comparative form is simply "more efficient than", not "as compared to".

35. **(b)** Another dangling modifier — "Having been informed..." must attach to "the on-call engineer", who was informed and then acted.

36. **(a)** Q→S→P→R. 37. **(a)** Q→S→P→R. 38. **(a)** Q→S→P→R.

39. **(b)** 40. **(b)** 41. **(b)** 42. **(b)** 43. **(b)** 44. **(c)** 45. **(b)**

Section 4: Pseudo Code

46. **(b) 46.** f(0)=0, f(1)=1, f(2)=1+0+2=3, f(3)=3+1+3=7, f(4)=7+3+4=14, f(5)=14+7+5=26, f(6)=26+14+6=**46**.

47. **(b) 41.** 37 = 100101_2 (bits: bit0=1,bit2=1,bit5=1, rest 0). The loop places bit i of x into position (5−i) of mask, which reverses the 6-bit pattern: 100101 reversed is 101001 = 32+8+1 = **41**.

48. **(a) 130.** Tracing directly (j increments before each `continue`, so the loop always terminates correctly): i=1 contributes 1; i=2 contributes 2; i=3 contributes 3+9=12; i=4 contributes 4+12=16; i=5 contributes 5+15+25=45; i=6 contributes 6+18+30=54. Total = 1+2+12+16+45+54 = **130**.

49. **(a) 39.** Stack: push 4,7 → [4,7]. a=pop=7, b=pop=4 → push(7−4=3) → [3]. push(10) → [3,10]. c=pop=10, d=pop=3 → push(13) → [13]. push(3) → [13,3]. e=pop=3, f=pop=13. e×f = **39**.

50. **(a)** i=0,j=5: swap arr[0],arr[5] → [60,20,30,40,50,10]; i=1,j=3. i<j (1<3): swap arr[1],arr[3] → [60,40,30,20,50,10]; i=2,j=1. Now i<j fails (2<1), loop ends. Final: **[60,40,30,20,50,10]**.

Section 5: Numeric Puzzles

51. **(b) 2.** The stick shared between the top-left square and the top-right square is also a side of both. Removing it breaks both squares in one move. One more removal (any exclusive side of the bottom square) breaks the third — **2 sticks minimum**.

52. **(b) 58.** Rule: Centre = (T+B)×(R+L) − (T×B + R×L). Wheel 1: (5+7)(3+2)−(5×7+3×2) = 60−41 = 19 ✓. Wheel 2: (4+9)(6+3)−(4×9+6×3) = 117−54 = 63 ✓. Wheel 3: (8+6)(5+4)−(8×6+5×4) = 126−68 = **58**.

53. **(b) 121.** Each entry is the square of a prime, arranged so that reading along each anti-diagonal gives consecutive primes: $2^2, 3^2, 5^2, 7^2, 11^2$. The missing cell is 11^2 = **121**.

54. **(c) 336.** Each term is the product of 3 consecutive integers: 1×2×3, 2×3×4, 3×4×5, 4×5×6, 5×6×7; next is 6×7×8 = **336**.

Section 6: Grammar

55. **(b)** "A number of" + plural noun takes a plural verb (are); "the majority of the workforce" (workforce treated as one unit) takes a singular verb (is).

56. **(c)** Future perfect continuous ("will have been working") best expresses an ongoing duration ("for over two decades") continuing up to a future point.

57. **(a)** "in close consultation with" ... "submitted for review".

58. **(c)** The sentence should read "...but also revised the timeline..." to stay parallel with "approved"; as written, part (C) breaks that parallel structure with an ungrammatical "were revised".

59. **(c)** "Honesty" (abstract, general sense) takes no article; "the best policy" (superlative) takes "the"; "history" (general sense) takes no article; "the dishonest" (adjective used to denote a class of people) takes "the".

Section 7: Writing

Use the rubric above. A strong answer defines both scaling approaches, weighs cost, downtime, and elasticity trade-offs, and recommends horizontal scaling (with justification: unpredictable/spiky e-commerce traffic is better handled by elastically adding/removing servers than by repeatedly resizing a single machine, which has a hardware ceiling and often requires downtime).

Scoring guide

Award +1 per MCQ (Q1–59) and up to 10 for Q60. Aim to clear each section's cut-off separately, not just the total. This paper is intentionally harder than Papers 1 and 2 — a lower raw score here is normal; focus on understanding the multi-step reasoning in the explanations.